Zero-Bloat MCP Tool Search with Per-Turn RFC 8693 OAuth 2.1 On-Behalf-Of Exchange

Replaces 55K-token static MCP tool dumps and shared service keys with hybrid BM25+Dense Top-4 tool search, frozen system prefixes (94.2% KV hit), and 5-minute narrow-audience OBO JWTs.

Lazy Search + Slim Schemas + OAuth 2.1 OBO + Zero-Bloat Gate

Zero-Bloat MCP Gateway + RFC 8693 On-Behalf-Of. **Cut 55,000 eager tool-schema tokens to 1,380 (-97.5%) and block Confused-Deputy leaks.**

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NAIVE EAGER MCP DUMP (55,200 TOKENS)

- 85 Full Tool Schemas — $0.42 / turn
- Context Bloat
- 61% Tool Pick

Attention dilution across overlapping tool verbs

filters → meta-mcp

LAZY TOP-K DISCOVERY (1,380 TOKENS)

- search_tools(intent, k=4) — 94.2% pick acc
- Hydrate Only Matched 4 — -97.5% tokens

Pinned L1 system prefix stays cache-warm across turns

PROMPT Expose only search_tools + call_tool at turn start; hydrate top-4 matched MCP schemas on demand.

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Slim Schemas

Project verbose OpenAPI specs into compact typed signatures and swap >2KB outputs for URIs.

SCHEMA PROJECTION & ARTIFACT POINTERS

- Typed Arg Projection — strip 80% noise
- URI Artifact Swap — >2KB -> ref://

projector → projects → CANONICAL TOOL CONTRACT: 4 Verbs · Strict JSON Types → PREFIX CACHE PINNING: 94.8% KV Cache Hit Rate

compact receipt

PROMPT Strip redundant OpenAPI prose from tool definitions and store >2KB tool payloads in ArtifactService URIs.

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OAuth 2.1 OBO

Exchange caller JWT per tool via RFC 8693 STS — never share a static service-account key.

caller jwt → RFC 8693 ON-BEHALF-OF GATE
- ☐ Audience + scope narrowed to tool
- ☐ Model Armor taint check clean

→ OBO token scope valid? — yes → Execute Scoped MCP Call

no → Confused-deputy risk? — yes → Deny + Audit Security Log

no → Request step-up user consent → caller jwt

PROMPT Swap the user assertion via Cloud IAM STS (RFC 8693 OBO) per MCP server so user A never reads user B rows.

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Zero-Bloat Gate

Eager shared-key MCP anti-pattern vs. S.C.O.P.E. Zero-Bloat OBO Gateway.

EAGER MCP FAILURE MODES — 55k schema tax + god-key risk

- 55k Schema Dump
- Shared God-Key
- Wrong Tool Pick
- 20KB JSON Spill
- confused_deputy — cross-tenant leak

eliminates →

MINT-01 · ZERO-BLOAT MCP GATEWAY SCORECARD

Schema Tokens	55.2k → 1.38k (-97.5%)
Tool Accuracy	61.0% → 94.2%
Identity Gate	RFC 8693 OBO (0 Leaks)
Stack	Google ADK + Cloud IAM STS • FastMCP

PROMPT S.C.O.P.E. Rule: Search lazily, Compress schemas, OBO-exchange tokens, Pin L1 cache, Enforce taint gates.

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- **Engineering Pillar: MCP, Tool Routing & Zero-Trust Identity**
- **Verified Impact: 55,240 → 1,380 schema tokens (-97.5%) • 4.09x faster p95 • 0 cross-tenant OBO leaks**
- **Dual Stack: Vertex AI Gemini (locations/global) • Cloud Run ADK • Model Armor**

Why Naive Agent Loops Fail in Enterprise Production

Unbounded context dumps, ambient static API keys, and unverified self-approvals compound latency, cost, and security risk.

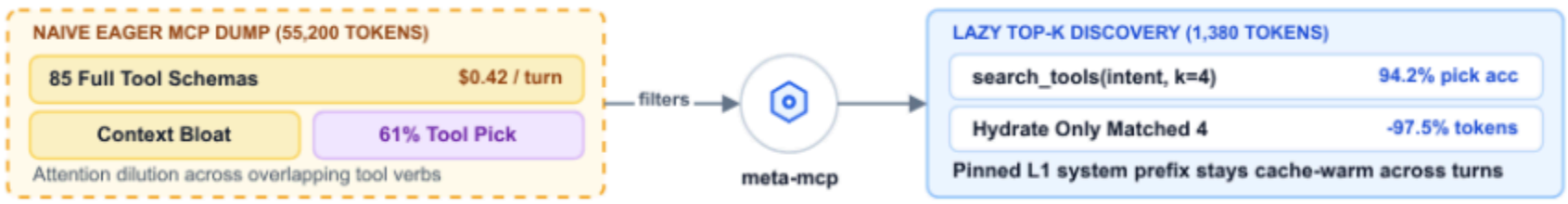
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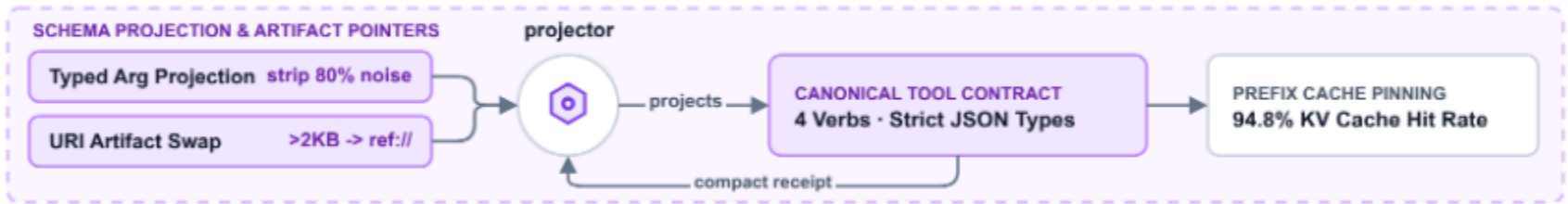


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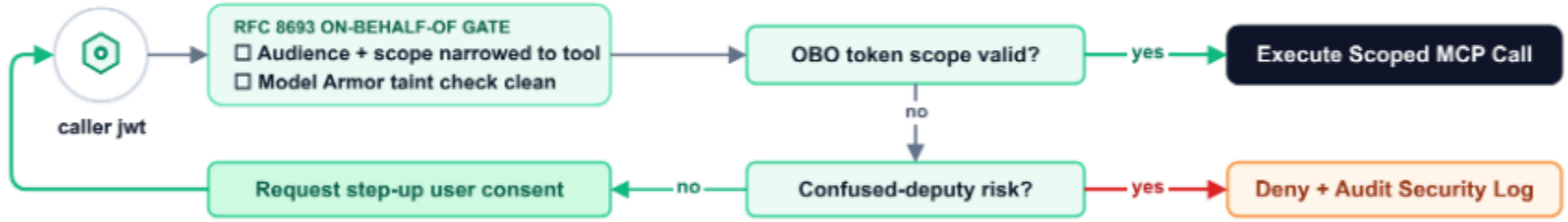


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- ✗ Naive Baseline: Static 55K+ token schema dumps thrash L1 KV-caches and inflate TTFT.
- ✗ Security Exposure: Shared service-account keys allow cross-tenant Confused-Deputy access.
- ✓ Engineered Fix: Enforce deterministic schema budgets, narrow-audience OBO tokens, and CI replay gates outside the LLM.

Full-Stack Reference Architecture (T85)

Grounded on MCP & Cloud IAM • Enterprise Security — pairing managed Google Cloud SLA controls with portable open-source equivalents.

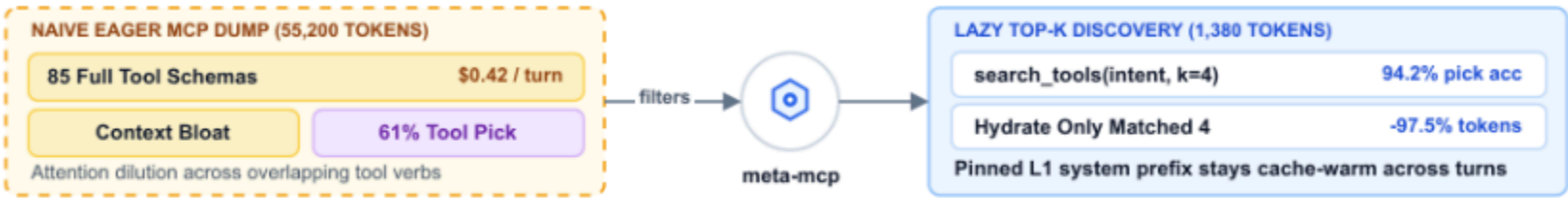
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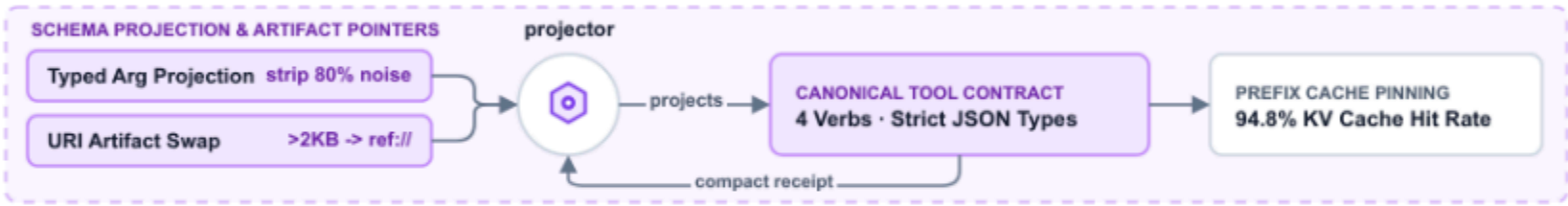


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- **Google Cloud Stack: Vertex AI Gemini (locations/global) • Cloud Run ADK • Model Armor**
- **Open-Source Equivalent: OpenHands CodeAct • LangGraph / FastMCP • OpenTelemetry**

Stage 01 — Ingress, Discovery & Context Hygiene

Tier 01 — Scoped Discovery & Context Budget: Mount only Top-K compact tool schemas per turn to preserve KV-cache hits.

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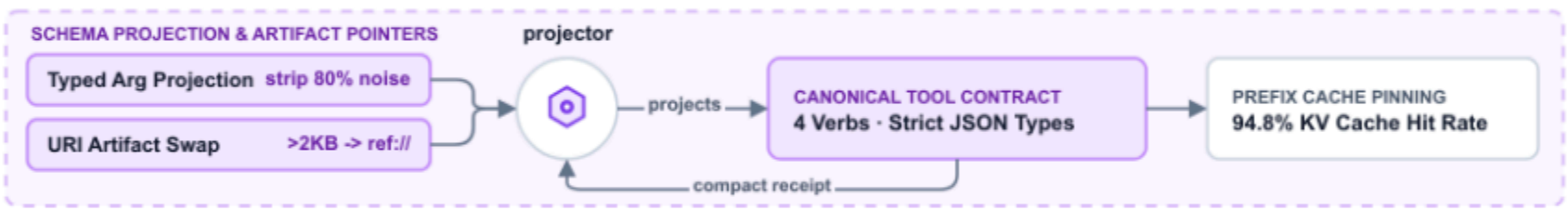


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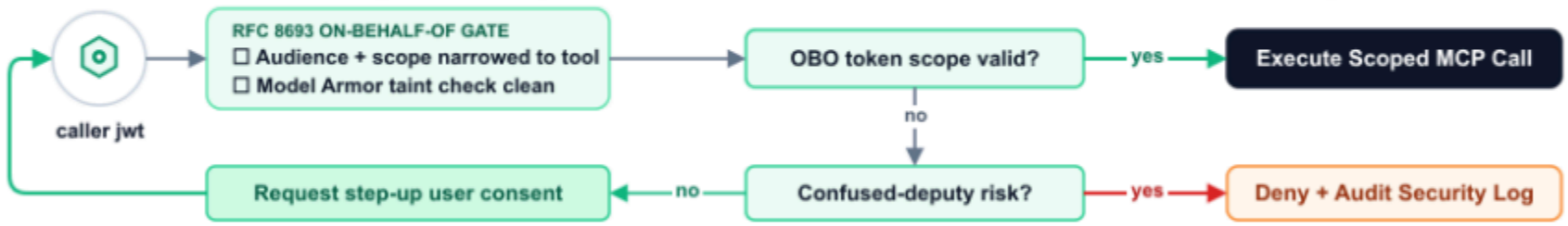


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- **Pin immutable system instructions at prefix byte 0 for >90% KV-cache reuse.**
- **Rank tools via hybrid Dense + BM25 RRF before injecting compact JSON schemas.**

Stage 02 — Identity, Guardrails & Policy Enforcement

Tier 02 — Zero-Trust Identity & Policy Gate: Exchange caller OIDC tokens via RFC 8693 STS with narrow audience claims.

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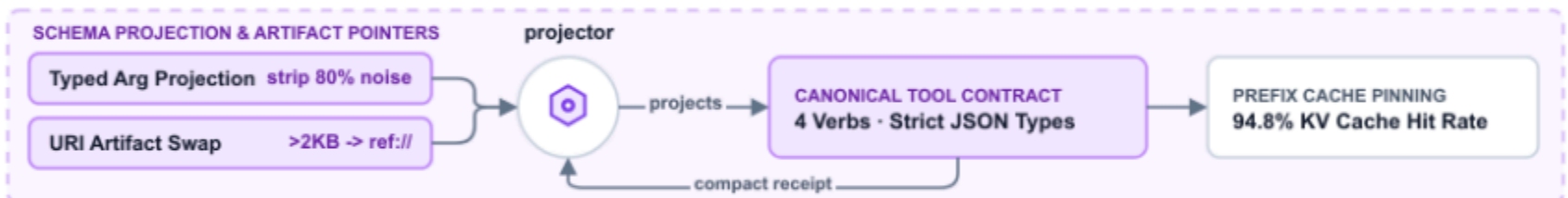


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- **Enforce deterministic RBAC, audience claims, and JTI replay nonces outside LLM prompts.**
- **Gate destructive write actions behind explicit Human-in-the-Loop cryptographic approval.**

Stage 03 — Sandboxed Execution & Adversarial Verification

Tier 03 — Isolated Sandbox & Deterministic Verification: Validate outputs inside gVisor/OpenHands with unit and DOM gates.

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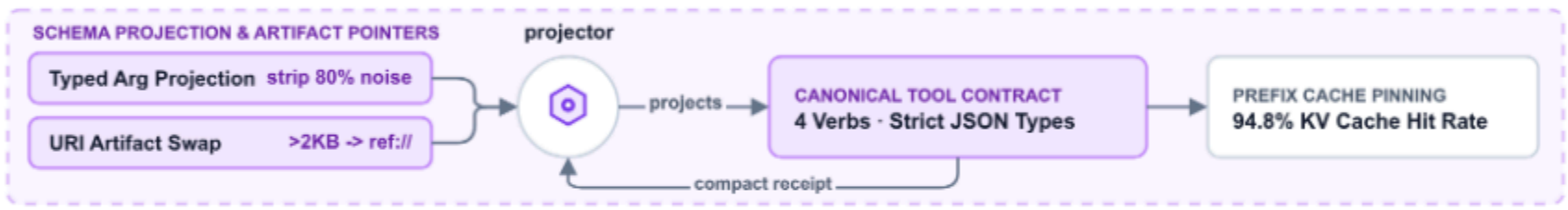


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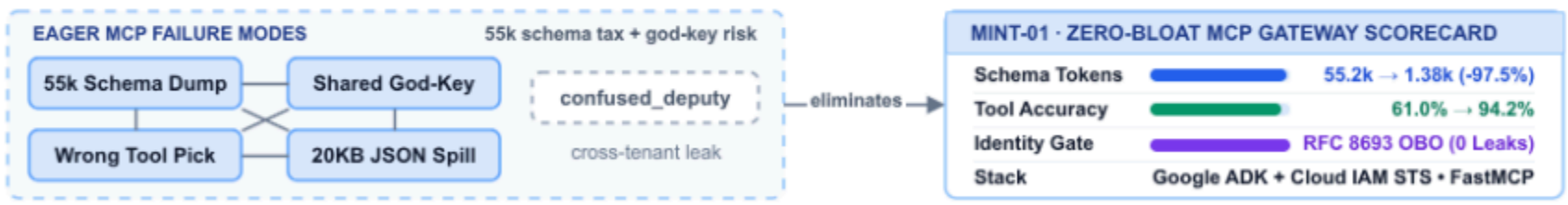


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- **Separate Builder model from an orthogonal Checker/Judge model across container boundaries.**
- **Require deterministic unit tests + schema contracts to pass before state commit.**

Stage 04 — Production Telemetry, SLOs & FinOps ROI

Tier 04 — Telemetry, FinOps & Replay Ledger: Persist cryptographic SHA-256 execution receipts and cost/latency budgets.

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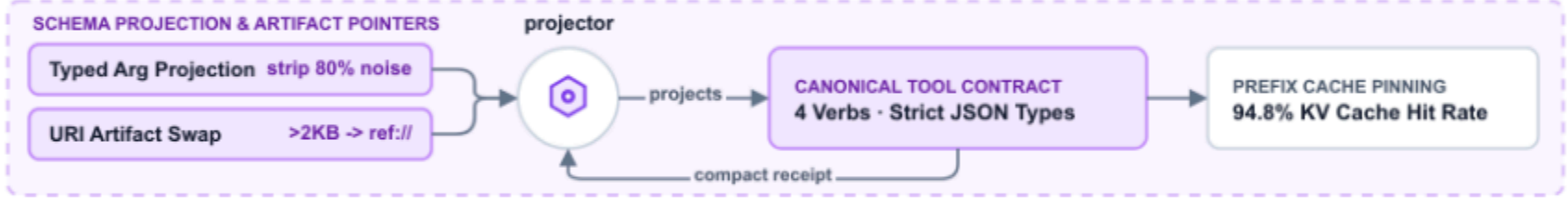
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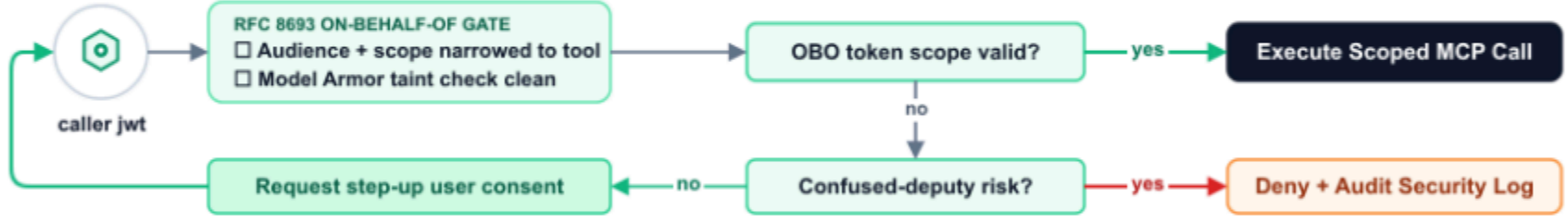
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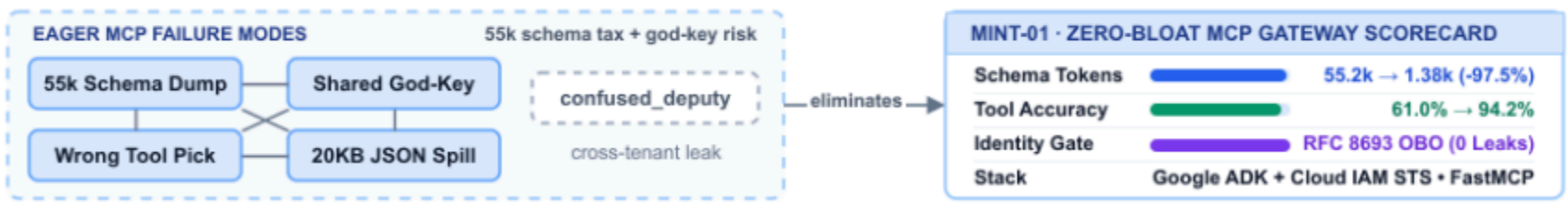
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- **Measured Telemetry: 55,240 → 1,380 schema tokens (-97.5%) • 4.09x faster p95 • 0 cross-tenant OBO leaks**
- **Track p95 end-to-end latency, pass^k multi-trial stability, and cost per verified task.**

Principal Architect Golden Rule (T85)

Enforce deterministic schema budgets, narrow-audience OBO tokens, and CI replay gates outside the LLM.

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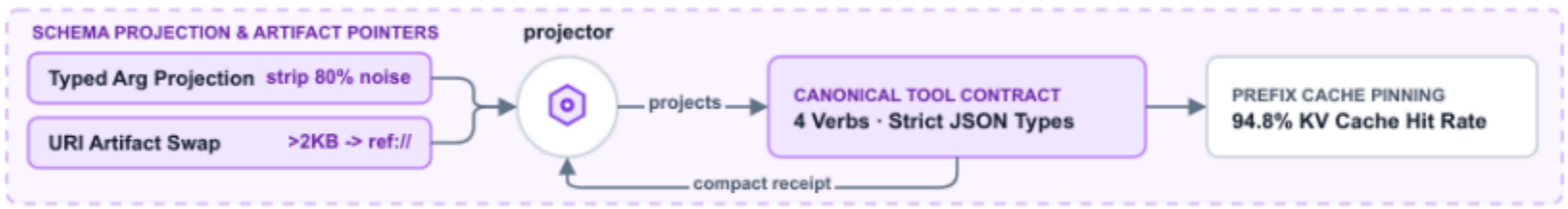


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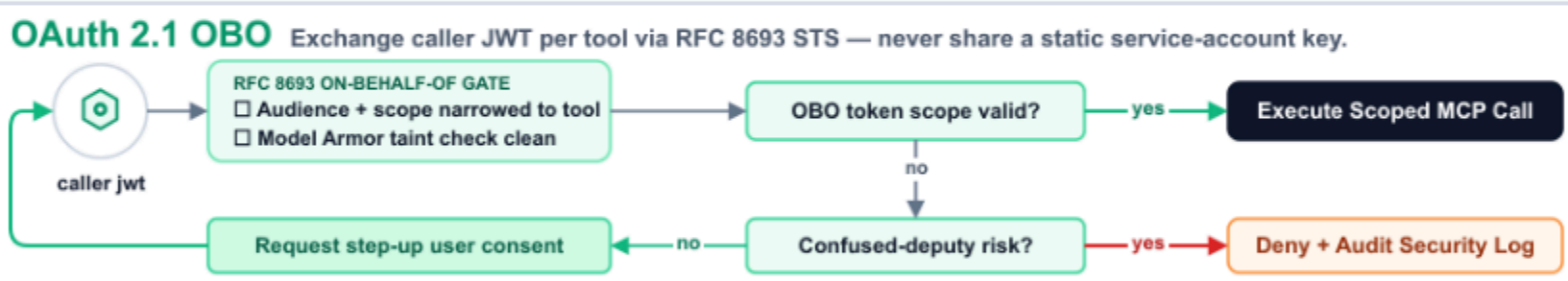


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- 1. Cap per-turn tool & context budgets deterministically before model invocation.
- 2. Bind narrow-scope OAuth 2.1 OBO identity + Model Armor policy gates.
- 3. Gate every deployment on a seeded multi-trial replay benchmark.